



ÉCOLE POLYTECHNIQUE DE SOUSSE

DHCP Server Deployment on Windows Server 2019

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Chapitre 1

DHCP Server Deployment on Windows Server 2019

1.1 Objective

This lab focused on deploying and configuring a **DHCP (Dynamic Host Configuration Protocol)** server within an isolated VMware lab environment to automate IP address assignment for Windows clients. The DHCP server was also integrated as a member server of an Active Directory domain (poly.local) and configured to act as the domain's DNS server, reflecting a typical enterprise dual-role deployment.

1.2 A. Test Environment & Hardware Specifications

Element	Configuration
Hypervisor	VMware Workstation (LAN Segment mode)
Server (SRV-DC1)	Windows Server 2019, 2 vCPU, 4 GB RAM
Client (CL1)	Windows 10, 2 vCPU, 4 GB RAM
Virtual Network	Isolated LAN Segment : LAN_DHCP_TEST

TABLE 1.1 – Virtual lab environment configuration

1.3 B. Network Isolation Setup

A VMware LAN Segment named LAN_DHCP_TEST was created and assigned to both VMs (server and client). Unlike a NAT or bridged network, a LAN Segment functions as a private virtual switch, isolating traffic between the two machines from the host network and external interference — a standard approach for building safe, contained test labs before deploying changes to production.

1.4 C. Server Configuration (SRV-DC1)

- **Static IP address** : 10.10.10.10 /24
- **Server name** : SRV-DC1
- Active Directory Domain Services (AD DS) deployed, creating the domain poly.local
- DHCP Server role installed via Server Manager
- DHCP server authorized within Active Directory (a mandatory step in a domain environment — an unauthorized DHCP server is ignored by domain-joined clients as a rogue-server security safeguard)

At this stage, the server operates as both DNS and DHCP, consolidating name resolution and IP address management on a single host — simplifying internal service discovery and administration.

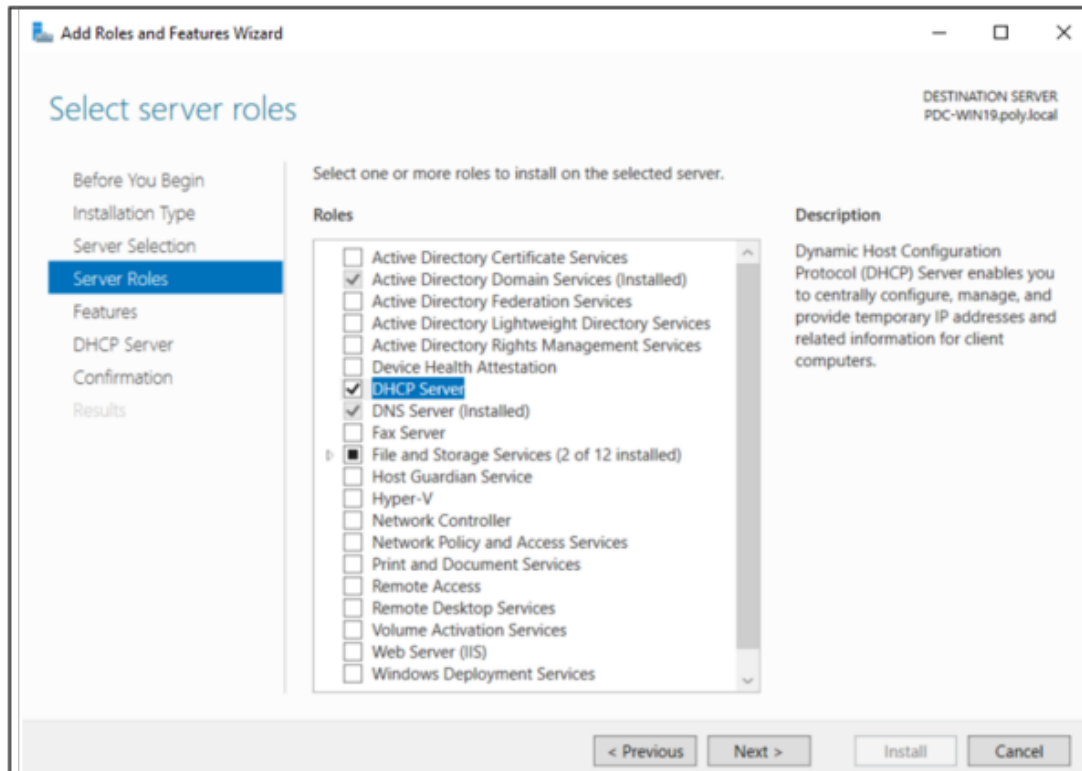


FIGURE 1.1 – Server role selection : DHCP Server and DNS Server installation via Server Manager

1.5 D. DHCP Scope Configuration (10.10.10.0/24)

An IPv4 scope named SCOPE-10.10.10.0-24 was created with the following parameters :

- Address range : 10.10.10.1 → 10.10.10.254
- Exclusion range : 10.10.10.1 → 10.10.10.10 (reserved for static infrastructure addresses, e.g., the server itself)
- Lease duration : 5 days, 12 hours, 30 minutes
- Router (default gateway) option (003) : 10.10.10.1
- DNS server option (006) : 10.10.10.10
- Domain name option (015) : poly.local

Excluding the low end of the range from the dynamic pool is a common best practice to avoid IP conflicts with statically assigned infrastructure devices. The scope was then activated, enabling it to lease addresses to clients.

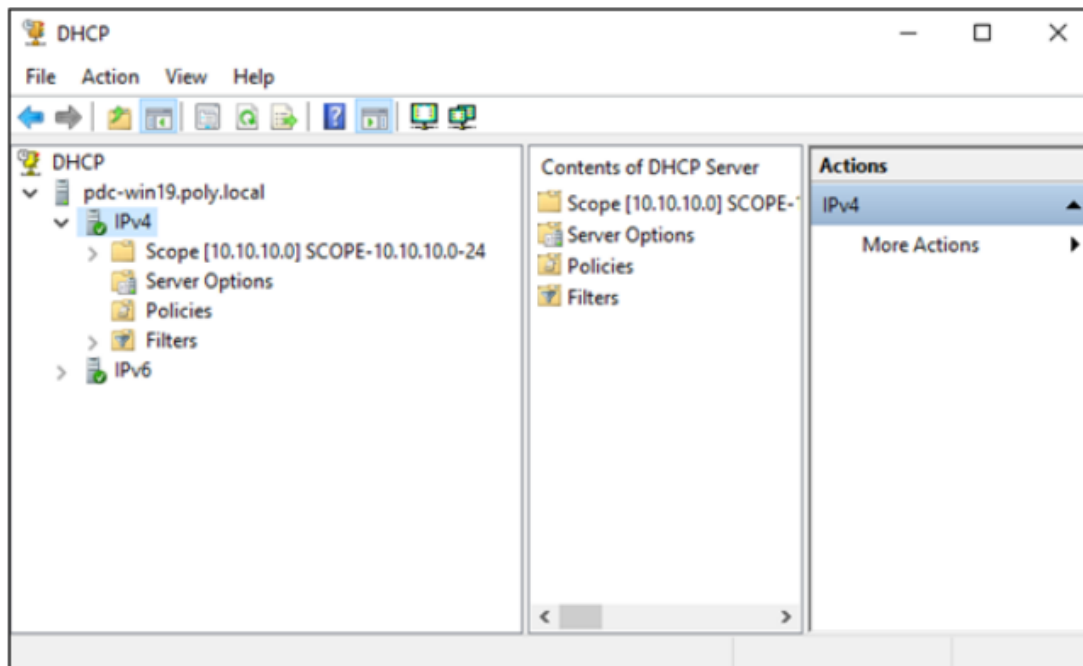


FIGURE 1.2 – DHCP console showing the active scope and server options

1.6 E. Client Configuration (CL1)

The Windows 10 client's network adapter was set to "Obtain an IP address automatically" and "Obtain DNS server address automatically", delegating configuration entirely to the DHCP server rather than relying on static settings.

After running `ipconfig /renew`, the client received the following lease :

TABLE 1.2 – Network Configuration

Parameter	Value Assigned
IPv4 Address	10.10.10.11
Default Gateway	10.10.10.1
DNS Server	10.10.10.10
DNS Domain	poly.local

1.7 F. Connectivity & Name Resolution Testing

Connectivity and DNS functionality were validated using standard command-line diagnostic tools :

- `ping 10.10.10.10` → successful reply (0% packet loss), confirming basic IP-level connectivity to the server.
- `ping poly.local` → successful resolution and reply, confirming the domain name correctly resolves via DNS to the server's IP.
- `nslookup poly.local` → returned the correct name-to-IP mapping, validating DNS zone configuration, despite an initial request timeout (a transient behavior addressed by the second query showing correct resolution).

```
C:\Users\administrator>ping 10.10.10.10

Pinging 10.10.10.10 with 32 bytes of data:
Reply from 10.10.10.10: bytes=32 time<1ms TTL=128
Reply from 10.10.10.10: bytes=32 time=1ms TTL=128
Reply from 10.10.10.10: bytes=32 time<1ms TTL=128
Reply from 10.10.10.10: bytes=32 time<1ms TTL=128

Ping statistics for 10.10.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

```
C:\Users\administrator>ping poly.local

Pinging poly.local [10.10.10.10] with 32 bytes of data:
Reply from 10.10.10.10: bytes=32 time<1ms TTL=128
Reply from 10.10.10.10: bytes=32 time<1ms TTL=128
Reply from 10.10.10.10: bytes=32 time=1ms TTL=128
Reply from 10.10.10.10: bytes=32 time<1ms TTL=128

Ping statistics for 10.10.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

```
C:\Users\administrator>nslookup poly.local
DNS request timed out.
    timeout was 2 seconds.
Server:    UnKnown
Address:   10.10.10.10

Name:     poly.local
Address:  10.10.10.10
```

FIGURE 1.3 – Connectivity validation via ping to server IP and domain name

1.8 G. Domain Join

CL1 was joined to the poly.local domain using the poly Administrator account. Following a restart, the domain connection was verified, and the client's DHCP lease correctly appeared in the server's DHCP console — confirming end-to-end integration between DNS, DHCP, and Active Directory.

1.9 H. DHCP Reservation (Static Mapping via DHCP)

A **DHCP reservation** was configured for client CL1 to guarantee it consistently receives the same IP address despite using dynamic allocation — a common practice for servers, printers, or infrastructure devices that require a predictable address without manual static configuration :

- **Reservation name** : CL1
- **Reserved IP address** : 10.10.10.20
- **MAC address** : 00-0C-29-B9-91-C4
- **Supported types** : DHCP
-

The client's lease was then refreshed using `ipconfig /release` followed by `ipconfig /renew`, confirming it correctly received the reserved address (10.10.10.20) rather than a random address from the pool — validating the reservation was applied successfully.

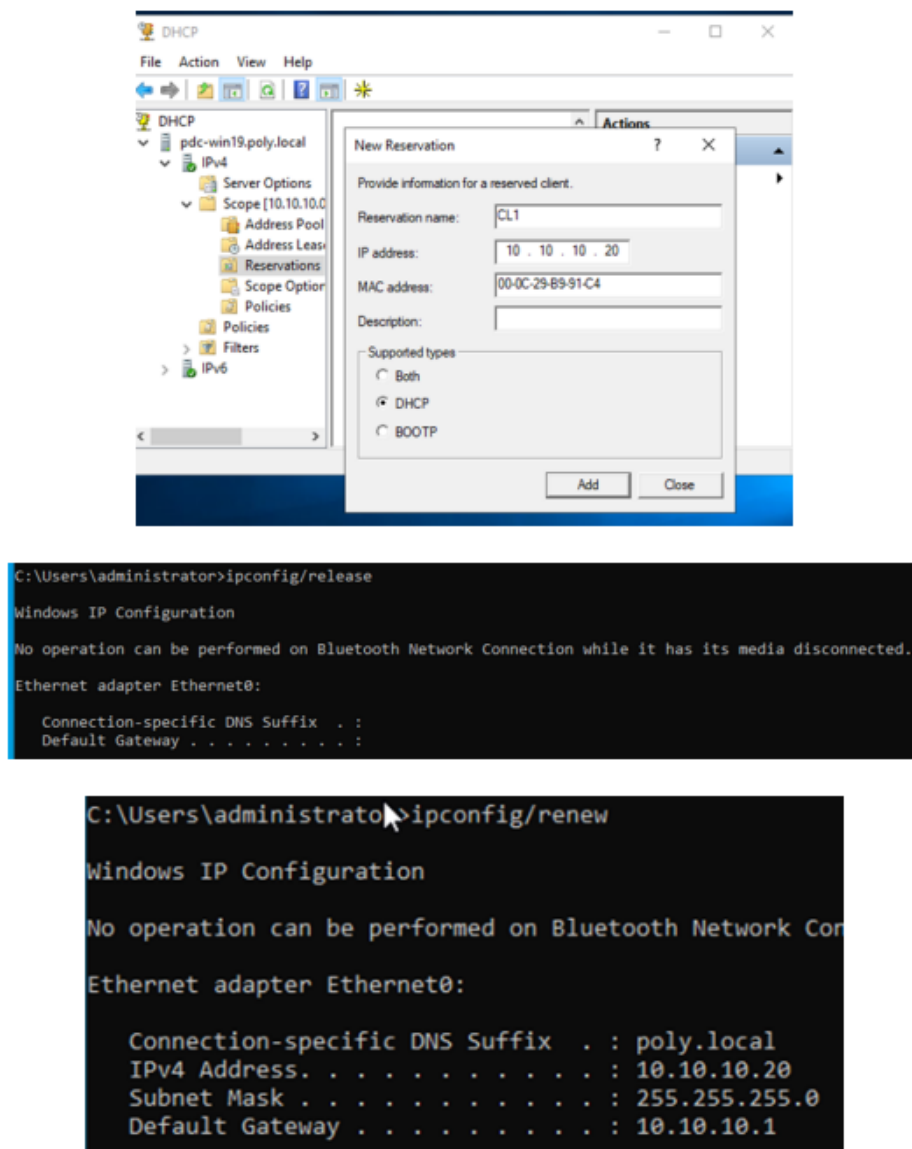


FIGURE 1.4 – IP lease renewal via `ipconfig /release` and `/renew`, confirming reserved address

1.10 Key Takeaways

This lab provided practical, end-to-end experience deploying and validating a DHCP infrastructure integrated with Active Directory Domain Services in an isolated test environment. Key outcomes :

- Successfully deployed and authorized a DHCP server within an AD DS domain, understanding the security rationale behind server authorization
- Configured a dual-role DNS/DHCP server, a common consolidation pattern in small-to-mid-sized enterprise networks

- Applied scope design best practices, including exclusion ranges and lease duration tuning
- Implemented and validated a DHCP reservation, demonstrating the ability to provide predictable addressing for critical devices without manual static configuration
- Verified the full IP provisioning and name-resolution chain using standard network diagnostic tools (ping, nslookup, ipconfig)

This exercise reinforced core competencies in network services administration directly applicable to enterprise infrastructure and IT operations roles — including troubleshooting client connectivity issues at the DHCP/DNS layer, a frequent real-world support and security concern (e.g., rogue DHCP detection).

